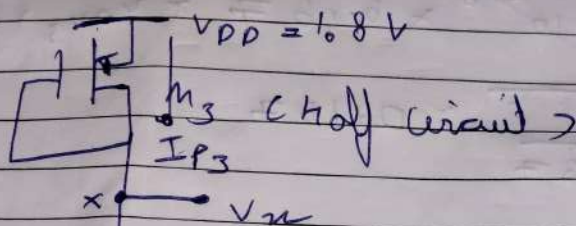


DC Analysis :-

Symmetry



$$I_{D1} = \frac{I_{SS}}{2} = I_{D2}$$

$$I_{DT} = I_{D3}$$

$$I_{D1} = I_{D3} = I_{SS}$$

$$I_{SS} = \frac{1}{2} \times \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{th})^2$$

$$0.8V = V_{GS}$$

$$I_{SS} = \frac{1}{2} \times 420 \times 10^{-6} \times 3.75 \times 10^{-6} \times (0.8 - 0.65)^2$$

$$I_{SS} = \frac{1}{2} \times 420 \times 10^{-6} \times 3.75 \times 10^{-6} \times (0.8 - 0.65)^2 = 31.6 \mu A$$

$$I_{D1} = 15.8 \mu A = I_{D3}$$

$$V_P = V_{D5} \Rightarrow I_{D1} = \frac{1}{2} \times \mu_n C_{ox} \frac{W}{L} \times (V_{OV1})^2$$

$$V_{ov1} = \sqrt{\frac{2 I_{D1}}{\mu_n C_{ox} \frac{W}{L}}} = \sqrt{\frac{15.08 \times 10^{-6}}{42 \times 10^{-5} \times 6.67}} \approx \sqrt{9.80 \times 10^{-1}} = 0.106$$

$$\therefore V_{GS} - V_{thn} = 0.106 \text{ V} \Rightarrow V_{GS} = 0.756 \text{ V}$$

$$V_{G1} - V_S = 0.756 \Rightarrow V_S = 1.0 - 0.756 = V_P$$

$$V_P (\text{tail node}) = 0.345 \text{ V}$$

$$I_{P3} = \frac{1}{2} \times \mu_p C_{ox} \frac{W}{L} \times (V_{OV3})^2$$

$$I_{P3} = I_{P3} = \frac{1}{2} \times 168 \times 10^{-6} \times 6.67 \times (V_{OV3})^2$$

$$V_{OV3} = \sqrt{\frac{31.6}{168 \times 6.67}} \approx 0.167$$

$$V_{SG} - |V_{thp}| = 0.167 \Rightarrow V_{SG} = 0.167 + 0.6 = 0.767 \text{ V}$$

V_{out1}

$$V_{in} = V_{DD} - V_{SG3}$$

$$V_{in} = 1.0 - 0.763 \Rightarrow 1.037 \text{ V}$$